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Generalized Biot's theory and Mandel's problem of multiple-porosity and multiple-permeability poroelasticity

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Abstract This paper finds in Biot's theory of poroelasticity a complete and consistent extension to the general case of multiple-porosity and multiple-permeability, fluid-saturated, and linearly elastic media. The constitutive stress-strain relations for a medium identified with this extension are presented, and the coefficient matrix of mechanical properties associated with these relations is derived from the corresponding intrinsic properties of its single-porosity constituents. The closed form analytical solution to Mandel's problem is upgraded to the case being considered in this study. This problem addresses the transient consolidation of a porous elastic slab of rectangular geometry, when confined from the top and bottom. A numerical example solution for shale with laboratory setup of Mandel's problem is provided. Results are compared for the cases of single-, double-, and triple-porosity solutions.

1. Introduction

Since the advent of basic ideas in the theory of fluid seepage in fissured porous media [Barenblatt *et al.*, 1960; Warren and Root, 1963], much effort has been made to foster the dual-continuum method in mechanical modeling of materials with more than one distinguishable porous structures. The dual-diffusivity theory of Aifantis [1979, 1980a, 1980b] represents earlier investigations on the subject, with a focus on the species diffusion through a rigid solid matrix. More recently, a discrete representation of the theory has been published by Kalampakas and Aifantis [2012] by use of the graph theory. It has been shown that phenomenological coefficients of dual-diffusivity formulation at this level of investigation require some extra constraints, depending on geometry and graph topology, which cannot be predicted by a macroscopic approach. Another recent and interesting advancement on this subject is the *gradient nanomechanics* [Aifantis, 2011], which considers interactions between grain boundaries and the bulk phase. These interactions are presented in the form of extra gradient terms contributing to the conservation laws of the associated constitutive variables which govern the nanoscale behavior of the material.

Alternatively, the theory of dual-porosity and dual-permeability poroelasticity [Aifantis, 1986; Wilson and Aifantis, 1982; Beskos and Aifantis, 1986; Berryman and Wang, 1995] stems from Biot's theory of poroelasticity [Biot, 1941, 1955] to address the coupled fluid flow and deformation processes of such media. Early on, Lewallen and Wang [1998] developed an exact analytical solution for 1-D consolidation of a double-porosity column of rock. In that solution, the histories of two distinct pore pressures, as well as displacement of the column's top surface, were shown to be functions of the contrast between mechanical and flow properties of the primary and secondary porosities. More recently, the 2-D counterparts to this problem—known as Mandel-type problems of poroelasticity—have been successfully upgraded to the case of dual-porosity and dual-permeability materials [Nguyen and Abousleiman, 2010]. Further analytical solutions in this context arise from their applications in the oil and gas industry, such as the wellbore stability [Zhang *et al.*, 2003; Abousleiman and Nguyen, 2005] and compaction [Nguyen and Abousleiman, 2009] of naturally fractured reservoirs, or in biomechanics, including the dual-porosity poroelastic analyses of bone [Smit *et al.*, 2002], brain [Wirth and Sobey, 2009], and intervertebral disc tissues [Huyghe *et al.*, 2003].

Despite its elegant and natural approach in extending Biot's theory to account for two distinct pore structures in an elastic medium, the dual-porosity/dual-permeability model may fall short in capturing some crucial

aspects of flow and/or deformation in many fluid saturated porous materials. For this reason, generalizations to this model has been the subject of studies which consider either flow [Gwo *et al.*, 1995] or coupled flow-deformation processes [Bai *et al.*, 1993] in naturally fractured rocks. Shale is also known to comprise multiple-fracture networks ranging from the nanoscale to the sequence/systems tract scale in size and hydraulic conductivity [Slatt and Abousleiman, 2011]. Their very intricate pore structure has recently been the focus of investigations which aim to model the storage capacity and productivity [Dehghanpour and Shirdel, 2011; Deng *et al.*, 2011] or geomechanics [Abousleiman *et al.*, 2010] of oil- or gas-bearing shale formations.

Biomaterials such as bone and brain tissues are also well known for their multiple pore structure. For example, *Hierarchical poroelasticity* [Cowin *et al.*, 2009] and *Russian doll poroelasticity* [Gailani and Cowin, 2011] are the terms attributed to the specific models which address the interstitial fluid exchange between vascular and lacunar-canalicular porosities of the bone tissue. Likewise, the *multiple-network poroelasticity* [Tully and Ventikos, 2011] describes complexities associated with the brain tissue's mechanical deformation, arising from fluid transport and exchange through four porosity networks. These networks comprise the extracellular space (matrix), as well as three distinct porosities related to arteries, veins, and capillary blood vessels within the parenchyma.

This paper extends the existing theories of single- and dual-porosity, fluid-saturated, homogenous, isotropic, and elastic materials to account for higher levels of pore structure complexity, namely, the *multiple-porosity and multiple-permeability poroelasticity*. A number of conceptual models comprising groups of springs and fluid containers are presented in section 2. These models illustrate the poroelastic response of materials with a different numbers of coexisting porosity networks to external loading, through which a generalized constitutive stress-strain relationship for multiple poroelastic media is developed. Complete mechanical characterization of multiple-porosity mixtures is still an open problem in mechanics of porous media [Berryman, 2002]. This problem is addressed for a poroelastic mixture in Appendix A. For this purpose, a series of *thought experiments* are designated in order to find the mixture's mechanical properties as functions of similar intrinsic properties of its porous constituents.

Multiple-porosity and multiple-permeability theory, together with the effect of interporosity exchanges of pore fluid, are capable of handling related applied problems of rock mechanics. In order to show this capability, Mandel's problems are revisited in section 3. Owing to its generality and broad applications, a closed form analytical solution to this problem for the case of multiple-porosity and multiple-permeability poroelasticity is developed. Section 4 presents a numerical example of this solution for a triple-porosity sample of shale undergoing some confining stress with the boundary conditions adapted to Mandel's problem. Results are discussed and compared for the three cases of single-, double- and triple-porosity poroelastic models.

2. Poroelasticity

2.1. Single Porosity

Biot's theory of poroelasticity [Biot, 1941, 1955] incorporates the transient effects of pore fluid pressure in the analysis of the elastic skeleton's deformation. With constituents' compressibility taken into account, this theory manifests the volumetric deformations of solid and fluid phases through three independent material properties. The corresponding constitutive relations can be written in terms of drained matrix bulk modulus K , Skempton's pore pressure coefficient B [Skempton, 1954], and Biot-Willis coefficient α [Biot and Willis, 1957] as follows [Berryman, 2002]:

$$\begin{pmatrix} \varepsilon \\ \zeta \end{pmatrix} = \frac{1}{K} \begin{pmatrix} 1 & \alpha \\ \alpha & \alpha/B \end{pmatrix} \begin{pmatrix} \sigma \\ p \end{pmatrix} \quad (1)$$

where σ and p stand for the average normal stress (tensile positive) and excess pore pressure and ε and ζ denote the volumetric strain (Positive in expansion) and increment of fluid content (positive when fluid added to porous space), respectively.

An abstract presentation of poroelastic response to external stress is shown in Figure 1 in a method similar to schematics of Terzaghi's soil consolidation experiment [Wang, 2000]. In this setup, the fluid-filled container bounded by a piston and needle valve on top and together with a spring of stiffness k_s undergo a sudden compressive load F . The excess pore pressure instantaneously rises to $p(0^+)$, the occurrence of which is known

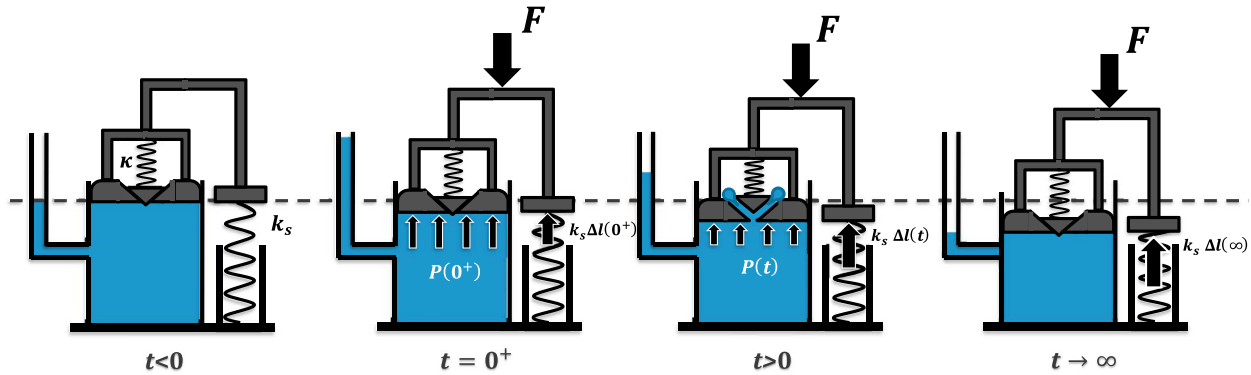


Figure 1. Poroelastic deformation is conceptually shown by combined response of the solid skeleton and pore fluid pressure $p(t)$. The hydraulic conductivity κ is represented by the needle valve and back spring on the top.

as Skempton's effect [Skempton, 1954] in poromechanics literature. Until its complete dissipation, the resisting force produced by this pressure gradually transfers to the solid skeleton. The rate of pore pressure dissipation in this process depends on both the hydraulic conductivity κ and effective stiffness of the porous mixture.

The quasi-static state of mechanical equilibrium in this setup defines its constitutive relation. This relation can be formulated as follows:

$$F = k_s \Delta l(t) + A p(t) \quad (2)$$

where A is the container's cross-section area. For a more concrete case which addresses deformations of porous elastic materials, the corresponding constitutive relation in three dimensions is written as follows [Rice and Cleary, 1976]:

$$\sigma_{ij} = (2G\epsilon_{ij} + \lambda\epsilon\delta_{ij}) - \alpha p\delta_{ij} \quad (3)$$

where G, λ are Lamé's constants. The minus sign for αp implies a tensile-positive convention for the stresses. The use of Kronecker delta function δ_{ij} in equation (3) is due to the fact that the pore fluid pressure does not affect the solid skeleton's shear deformations. Equations (1) and (3) complete the required constitutive relations for the general three-dimensional deformation of a homogenous, isotropic, and linearly poroelastic medium.

2.2. Dual Porosity and Dual Permeability

A class of related studies presents models for mechanical behavior of dual-porosity mixtures [Berryman and Berge, 1996; Berryman and Pride, 1998, 2002; Berryman, 2002]. Berryman's postulate for characterizing dual-porosity poroelastic materials incorporates a natural extension of equation (1) as follows:

$$\begin{pmatrix} \epsilon \\ \zeta_1 \\ \zeta_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \sigma \\ p_1 \\ p_2 \end{pmatrix} \quad (4)$$

This postulate inherits the assumption that both the mixture and its constituents would behave as *Gassmann materials*, rendering two distinct pore pressures p_1, p_2 corresponding to two coexisting porosity networks. On the basis of these models, expressions for *effective poroelastic properties* a_{ij} of such dual continua in terms of the *intrinsic properties* K_i, B_i, α_i of their constituents are derived [Berryman, 2002].

Within the same context, the abstract methodology taken in Figure 1 can be extended to the dual-porosity/dual-permeability case. Figure 2 portrays such generalization in a setup comprising a series of two springs, which represent the effective drained bulk modulus of the matrix. Besides, the two containers in each segment of Figure 2 denote two porosity networks with distinct fluid conductivities. These properties are represented by difference in stiffness of the back springs κ_1, κ_2 on top of the needle valves which reside

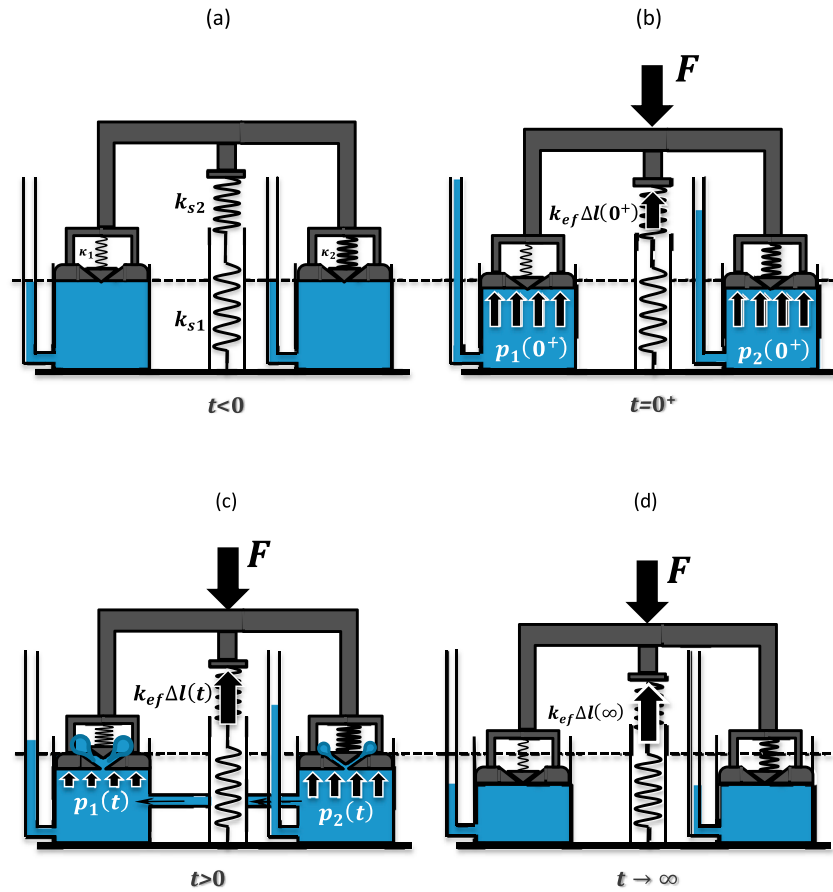


Figure 2. Conceptual presentation of transient dual-porosity/dual-permeability poroelastic deformation. (a) An elastic skeleton of equivalent spring constant $k_{ef} = (k_{s1}^{-1} + k_{s2}^{-1})^{-1}$ connected to two fluid containers through the loading fixture on top. (b) Sudden application of a compressive load on the system produces initial deformation $\Delta l(0^+)$ and pore pressure buildup of $p_1(0^+), p_2(0^+)$ within the containers. (c) Similar to the case of single-porosity poroelasticity, the pore pressures $p_1(t), p_2(t)$ dissipate with time through drainage from the needle valves on top of the containers. In this case, however, an interporosity flow $q_{2 \rightarrow 1} = \Gamma_{12}(p_2 - p_1)$ may also occur between the two porosity networks. (d) After complete dissipation of the pore pressures, the system returns to a new state of equilibrium characterized by the system's ultimate deformation $\Delta l(t \rightarrow \infty)$.

within the pressurizing piston. The two porous networks spontaneously respond to a sudden compressive force at $t = 0^+$. As the time goes on ($t > 0$), these containers allow for the fluid not only to drain from the top valves but also to exchange through their interconnecting Poiseuille tube.

We notice the quasi-static state of equilibrium in the system for $t > 0$. The constitutive relation for the setup shown in Figure 2 can be accordingly written as follows:

$$F = k_{ef}\Delta l(t) + A_1 p_1(t) + A_2 p_2(t) \quad (5)$$

where A_1, A_2 are the cross-section areas of the two containers. A natural extension to constitutive relation of dual-porosity poroelastic materials in three dimensions can be written as follows:

$$\sigma_{ij} = 2G\epsilon_{ij} + \lambda\epsilon\delta_{ij} - (\bar{\alpha}_1 p_1 + \bar{\alpha}_2 p_2)\delta_{ij} \quad (6)$$

where $\bar{\alpha}_1, \bar{\alpha}_2$ are the effective Biot-Willis coefficients of the mixture. More elaborate discussions on these properties and their derivation for the general case of multiple-porosity materials is given in the next section and Appendix A.

2.3. Multiple Porosity and Multiple Permeability

Following Berryman's postulate for characterizing dual-porosity poroelastic materials, a generalized form of equation (4) can be written as follows:

$$\begin{pmatrix} \varepsilon \\ \zeta_1 \\ \vdots \\ \zeta_N \end{pmatrix} = \begin{pmatrix} a_{11} & \cdots & a_{1(N+1)} \\ \vdots & \ddots & \vdots \\ a_{(N+1)1} & \cdots & a_{(N+1)(N+1)} \end{pmatrix} \begin{pmatrix} \sigma \\ p_1 \\ \vdots \\ p_N \end{pmatrix} \quad (7)$$

Equation (7) characterizes a multiple-porosity porous medium comprising N distinct porous continuum networks, with a porous matrix of specific volume $v_1 = V_1/V$ and intrinsic material properties K_1, B, α_1 , as well as $(N - 1)$ additional porous constituents embedded within the matrix, having volume fractions $v_i = V_i/V$ and intrinsic material properties K_i, B_i, α_i . Total normal stress (tensile positive) and the volumetric strain (positive in expansion) denote σ and ε , and p_i, ζ_i , respectively, denote the excess pore pressures and increment of fluid content (positive when fluid added to porous space) in pore space of the i th porosity network. Since a linearly poroelastic model is being considered, a thermodynamic energy density property $U = \frac{1}{2} (\sigma \varepsilon + \sum_{i=1}^N p_i \zeta_i)$ can be defined for this material. The continuity of mixed partial derivatives of U requires the coefficient matrix in equation (7) to be symmetric, i.e., $a_{ij} = a_{ji}$. Thus, in order to characterize an N porosity linear poroelastic continuum, only $\frac{(N+1)(N+2)}{2}$ independent coefficients a_{ij} are required to be determined from the intrinsic properties attributed to the individual networks. A thought experiment is introduced in Appendix A which incorporates independent adjustments of $\sigma, p_1, \dots, p_N$ on a multiple-porosity material. Through this experiment and its related derivations, expressions for a_{ij} are obtained in terms of K_i, B_i, α_i for the general case of multiple-porosity poroelastic materials.

Following some simple linear algebra, equation (7) can be modified to the following form:

$$\begin{pmatrix} \sigma \\ \zeta_1 \\ \vdots \\ \zeta_N \end{pmatrix} = \begin{pmatrix} \bar{K} & -\bar{\alpha}_1 & \cdots & -\bar{\alpha}_N \\ \bar{\alpha}_1 & \frac{1}{\bar{M}_{11}} & \cdots & \frac{1}{\bar{M}_{1N}} \\ \vdots & \vdots & \ddots & \vdots \\ \bar{\alpha}_N & \frac{1}{\bar{M}_{1N}} & \cdots & \frac{1}{\bar{M}_{NN}} \end{pmatrix} \begin{pmatrix} \varepsilon \\ p_1 \\ \vdots \\ p_N \end{pmatrix} \quad (8)$$

with a more familiar coefficient matrix, whose elements are functions of a_{ij} . The total N parameters α_i appearing in the first row of the coefficient matrix in equation (8) are defined as the effective Biot-Willis coefficients of the mixture. $\bar{M}_{11}$ to $\bar{M}_{NN}$ are the corresponding effective Biot's moduli, while the effect of excess pore pressure of one network on the fluid content increment of the other is represented by interporosity modulus $\bar{M}_{ij}$. We notice that equation (8) defines the *effective* properties of a multiple-porosity substance (e.g., $\bar{\alpha}_i, \bar{M}_{ij}$), as opposed to the intrinsic property of its individual porous components α_i, M_{ij} .

The sequence of conceptual models for the hierarchy of deforming multiple-porosity elastic media can be further extended to a more general case. Figure 3 presents a setup comprising a series of N springs which represent the drained bulk modulus of the matrix. This property is analogous to $\bar{K}$ in equation (8) establishing the effective bulk modulus of drained porous skeleton. In addition, N containers denote N porosity networks with distinct fluid deliverability as represented by the stiffness of back springs on the needle valves within the pressurizing piston. Owing to their different compressibilities, the porous networks respond differently to a sudden compressive force at $t = 0^+$. As the time goes on, these containers allow for the fluid not only to drain from the top valves but also to exchange through their interconnecting Poiseuille tubes. It should be noted that in this case, a combinatory of $\binom{N}{2} = \frac{N(N-1)}{2}$ exchange properties Γ_{ij} can be considered for the system.

The quasi-static state of equilibrium in the system for $t > 0$ defines its constitutive relation as follows:

$$F = k_{ef} \Delta l(t) + \sum_{i=1}^N A_i p_i(t) \quad (9)$$

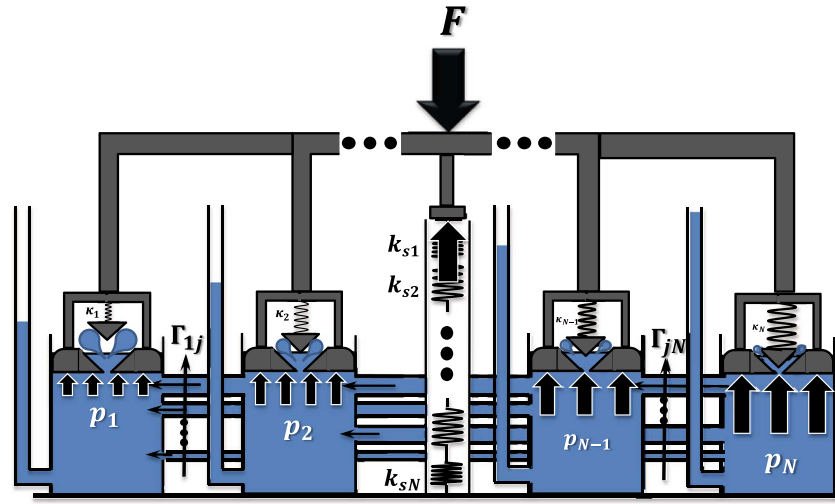


Figure 3. Conceptual presentation of multiple-porosity/multiple-permeability poroelastic deformation. An elastic skeleton of equivalent spring constant $k_{ef} = (\sum_{i=1}^N k_{si}^{-1})^{-1}$ connected to two fluid containers through the loading fixture on top. The pore pressures $p_1(t), \dots, p_N(t)$ dissipate with time through the needle valves on top of the containers, as well as interporosity flow $q_j \rightarrow_i = \Gamma_{ij}(p_j - p_i)$ between the porosity systems.

A natural extension to corresponding constitutive relation of multiple-porosity poroelasticity in three dimensions can be written as follows:

$$\sigma_{ij} = 2G\epsilon_{ij} + \lambda\epsilon\delta_{ij} - \sum_{i=1}^N \bar{\alpha}_i p_i \delta_{ij} \quad (10)$$

Equations (8) and (10) provide the required number of constitutive relations for deformation of a homogenous, isotropic, and linearly elastic medium with multiple-porosity networks. A little elaboration on describing various groups of plausible Biot-Willis coefficients, which arise from the related scales of investigations, seems necessary. At a micromechanics scale, the effective Biot-Willis coefficients $\bar{\alpha}_i$ of each porosity network, as appeared in equation (10), can be obtained from their intrinsic counterparts α_i as $\bar{\alpha}_i = (v_i/K_i)\alpha_i$. This relation can be deduced from manifestation of a series of thought experiments which independently control the confining stress and individual pore pressures of the multiple-porosity sample. Detailed description of these thought experiments, along with the related derivations, are presented in Appendix A.

Alternatively, and from a macroscopic level of investigation, one can define a global Biot-Willis coefficient $\bar{\alpha}$ for the entire fluid-saturated multiple-porosity matrix. Again, it is shown in Appendix A that $\bar{\alpha} = \sum_{i=1}^N \bar{\alpha}_i$. The permissible range of thermodynamically meaningful values for the macroscopic Biot-Willis coefficient is $0 \leq \bar{\alpha}_i \leq 1$ [Wang, 2000]. This constraint implies that $\bar{\alpha}_i$, appearing in the last term of equation (10) can, in a way, be interpreted as proper coefficients for a weighted average contribution of the pore pressures in distinct porosity networks to the constitutive effective stress relation of the multiple-porosity matrix.

2.4. Field Equations Featuring Interporosity Exchange Effects

The governing equations are herein outlined in a mixed formulation, that is, the volumetric strain ϵ and the excess pore pressure p are considered as the independent variables in the equilibrium and diffusion equations. The continuity of the fluid phase of porous matrix can be formulated through the following equation:

$$\frac{\partial \zeta_i}{\partial t} + \nabla \cdot \vec{q}_i = \sum_{j=1}^N \Gamma_{ij}(p_j - p_i) \quad i = 1, \dots, N \quad (11)$$

where the terms on the right-hand sides of equation (11) correspond to the sum of all interporosity flows from porosity network j to i . These fluxes are assumed to be proportional to their pore pressure contrast with a not necessarily symmetric coefficients matrix. However, without loss of generality, the assumption of

symmetry $\Gamma_{ij} = \Gamma_{ji}$ is implied through the derivations and solutions of this study. Equation (11), together with Darcy's law for each porosity network $\vec{q}_i = -\kappa_i \nabla p_i$ ($i = 1, \dots, N$), gives the following:

$$\frac{\partial \zeta_i}{\partial t} - \bar{\kappa} \nabla^2 p_i = \sum_{j=1}^N \Gamma_{ij} (p_j - p_i) \quad i = 1, \dots, N \quad (12)$$

where $\bar{\kappa}_i = v_i k_i / \mu$ is the hydraulic conductivity of i th porosity network. Substitution of the second to $(N+1)$ th row of the matrix relation from equation (8) into equation (12) yields

$$\bar{\alpha}_i \frac{\partial \varepsilon_i}{\partial t} + \sum_{j=1}^N \frac{1}{M_{ij}} \frac{\partial p_j}{\partial t} = \bar{\kappa}_i \nabla^2 p_i + \sum_{j=1}^N \Gamma_{ij} (p_j - p_i) \quad i = 1, \dots, N \quad (13)$$

The static equilibrium of a (poro)elastic medium can be mathematically written by the following equation:

$$\sum_{j=1}^N \frac{\partial \sigma_{ij}}{\partial x_j} = F_i, \quad i, j = 1, \dots, 3 \quad (14)$$

Under the condition of small displacements of material points, the strain tensor ε_{ij} in a continuum is obtained from kinematics of deformations as follows:

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (15)$$

The constitutive relation presented in equation (10), along with equation (15), can be substituted in equation (14) to give

$$G \nabla^2 \mathbf{u} + (\lambda + G) \nabla \varepsilon = \sum_{i=1}^N \bar{\alpha}_i \nabla p_i - \mathbf{F} \quad (16)$$

where $\mathbf{u}$ is the displacement vector and $\mathbf{F}$ is the body force.

3. Mandel's Problem

The two-dimensional Mandel's problem [Mandel, 1953] investigates consolidation of a porous rectangular strip of porous rock sealed and loaded from the top and bottom, while the fluid can freely drain from the sides. Likewise, a porous fluid-saturated sphere loaded on its permeable surface, from which the pore fluid can freely drain by a constant uniform pressure, is known as Cryer's problem [Cryer, 1963]. It has been shown through analytical solutions [Mandel, 1953; Abousleiman *et al.*, 1996] and laboratory experiments [Gibson *et al.*, 1963] that the pore fluid pressure in the interior of the sample jumps by an amount when the surface traction is applied and then continues to increase for some time before decaying. Being associated with poroelastic coupling, this phenomenon is called the Mandel-Cryer effect in poromechanics literature [Wang, 2000].

The industry practice of Mandel-type problems is quite common. Uniaxial and triaxial testing of porous rock specimens [Dickey *et al.*, 1968; Abousleiman and Cui, 1998] or investigations of sudden stress relief of a long core removed from subsurface wellbore [Wang, 2000] are examples of its laboratory applications. Further, it has been shown that if its boundary conditions are properly selected, a Mandel-type problem with hollow axisymmetric geometry can be used to analyze the compaction of producing reservoirs [Nguyen and Abousleiman, 2009]. Mandel's problem has also been the focus of biomechanical studies such as unconfined compression of articular cartilage [Armstrong *et al.*, 1984] and bone [Cowin and Mehrabadi, 1992].

Analytical solutions to Mandel-type problems with extension of Biot's theory to dual-porosity/dual-permeability Mandel-type problems have recently been developed [Nguyen and Abousleiman, 2010]. Using similar methodology and in line with formulations of section 2, solutions to the general case of multiple-porosity/multiple-permeability Mandel's problem is developed here. Figure 4 displays the geometry of this problem, along with boundary conditions adapted to this problem.

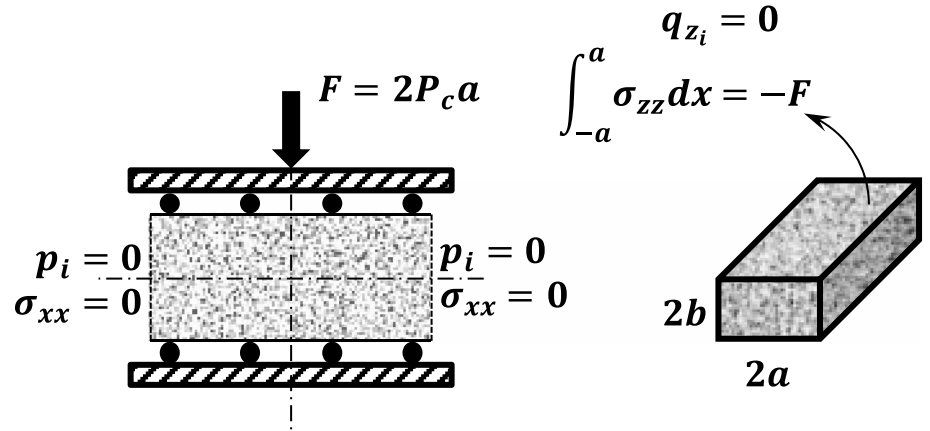


Figure 4. Geometry and boundary conditions of multiple-porosity/multiple-permeability Mandel's problem.

Because of the problem's symmetry in geometry and boundary conditions, the governing equations derived in section 2.4 need to be solved only for a quarter of the problem domain. Also, we notice that the displacement field features an irrotational distribution, that is, $\nabla \times \mathbf{u} = 0$. For such distribution it can be proved that $\nabla^2 \mathbf{u} = \nabla \varepsilon$. This condition simplifies equation (16) to

$$\nabla \varepsilon = \sum_{i=1}^N \bar{c}_{mi} \nabla p_i \quad (17)$$

which after integration yields

$$\varepsilon = \sum_{i=1}^N \bar{c}_{mi} p_i + f(t) \quad (18)$$

where the new parameters c_{m_i} are defined as

$$c_{m_i} = \frac{\bar{\alpha}_i(1 - 2\nu)}{2G(1 - \nu)} \quad (19)$$

Substitution of equation (18) into equation (13) gives

$$\left\{ \bar{\mathbf{S}} \frac{\partial}{\partial t} - \bar{\mathbf{\kappa}} \frac{\partial^2}{\partial x^2} + \bar{\mathbf{\Gamma}} \right\} \cdot \mathbf{p} = -\bar{\mathbf{a}} f'(t) \quad (20)$$

The symmetric matrices of generalized specific storage $\bar{\mathbf{S}}$ ($\bar{S}_{ij} = \bar{S}_{ji}$), hydraulic conductivity $\bar{\mathbf{\kappa}}$, and interexchange flow coefficient $\bar{\mathbf{\Gamma}}$, as well as the Biot-Willis coefficient $\bar{\mathbf{a}}$ and the pore pressure vector $\mathbf{p}$ are defined as follows:

$$\bar{\mathbf{S}} = \begin{pmatrix} \frac{1}{\bar{M}_1} + \bar{\alpha}_1 \bar{c}_{m1} & \frac{1}{\bar{M}_{12}} + \bar{\alpha}_1 \bar{c}_{m2} & \cdots & \frac{1}{\bar{M}_{1N}} + \bar{\alpha}_1 \bar{c}_{mN} \\ \frac{1}{\bar{M}_{12}} + \bar{\alpha}_2 \bar{c}_{m1} & \frac{1}{\bar{M}_{22}} + \bar{\alpha}_2 \bar{c}_{m2} & \cdots & \frac{1}{\bar{M}_{2N}} + \bar{\alpha}_2 \bar{c}_{mN} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\bar{M}_{1N}} + \bar{\alpha}_N \bar{c}_{m1} & \frac{1}{\bar{M}_{2N}} + \bar{\alpha}_N \bar{c}_{mN} & \cdots & \frac{1}{\bar{M}_{NN}} + \bar{\alpha}_N \bar{c}_{mN} \end{pmatrix} \quad (21)$$

$$\bar{\mathbf{\kappa}} = \begin{pmatrix} \bar{\kappa}_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \bar{\kappa}_N \end{pmatrix} \quad (22)$$

$$\mathbf{\Gamma} = \begin{pmatrix} \sum_{j=1}^N \Gamma_{1j} & -\Gamma_{12} & \dots & -\Gamma_{1N} \\ -\Gamma_{12} & \sum_{j=1}^N \Gamma_{2j} & \dots & -\Gamma_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ -\Gamma_{1N} & -\Gamma_{2N} & \dots & \sum_{j=1}^N \Gamma_{Nj} \end{pmatrix} \quad (23)$$

$$\bar{\mathbf{a}} = \begin{pmatrix} \bar{a}_1 \\ \vdots \\ \bar{a}_N \end{pmatrix} \quad (24)$$

$$\mathbf{p} = \begin{pmatrix} p_1 \\ \vdots \\ p_N \end{pmatrix} \quad (25)$$

Introducing the following dimensionless quantities

$$\bar{\kappa}_i = (\sum_{i=1}^N \kappa_i) \bar{\kappa}_i^*, \quad \bar{\Gamma}_{ij} = \frac{(\sum_{i=1}^N \kappa_i)}{a^2} \bar{\Gamma}_{ij}^*, \quad x = ax^*, \quad \bar{S}_{ij} = (\sum_{i=1}^N S_{ii}) \bar{S}_{ij}^*, \quad t = \frac{a^2 (\sum_{i=1}^N \bar{S}_{ii})}{(\sum_{i=1}^N \bar{\kappa}_i)} t^*$$

$$p_i = \frac{1}{(\sum_{i=1}^N \bar{S}_{ii})} p_i^*$$

equation (20) converts to

$$\left\{ \bar{\kappa}^* \frac{\partial^2}{\partial x^{*2}} - \bar{\mathbf{S}}^* \frac{\partial}{\partial t^*} - \mathbf{\Gamma}^* \right\} \mathbf{p}^* = \bar{\mathbf{a}} f'(t^*) \quad (26)$$

where the new matrixes and vectors $\bar{\kappa}^*$, $\bar{\mathbf{S}}^*$, $\mathbf{\Gamma}^*$, $\mathbf{p}^*$, $\bar{\mathbf{a}}$ are redefined accordingly. Laplace transformation of equation (26) on the time variable t results in the following equation:

$$\left\{ \kappa \frac{\partial^2}{\partial x^2} - (s\mathbf{S} + \mathbf{\Gamma}) \right\} \tilde{\mathbf{p}} = \mathbf{a} s \tilde{f} \quad (27)$$

where the asterisks associated with dimensionless parameters are dropped for simplicity in notation. The transformed variable in Laplace space is denoted by s , and the tilde $\sim$ on top identifies a transformed function. The solution of equation (27) is found from theory of system of ordinary differential equations (ODEs) and can be stated in the following form [Edwards and Penny, 1996]:

$$\tilde{\mathbf{p}}(r) = \begin{pmatrix} \chi_{11} & \chi_{12} & \dots & \chi_{1N} \\ \chi_{21} & \chi_{22} & \dots & \chi_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \chi_{N1} & \chi_{N2} & \dots & \chi_{NN} \end{pmatrix} \left\{ \begin{pmatrix} A_1(s) \cosh(x\sqrt{\lambda_1}) \\ \vdots \\ A_N(s) \cosh(x\sqrt{\lambda_N}) \end{pmatrix} + \begin{pmatrix} B_1(s) \sinh(x\sqrt{\lambda_1}) \\ \vdots \\ B_N(s) \sinh(x\sqrt{\lambda_N}) \end{pmatrix} \right\} + \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix} \tilde{f} \quad (28)$$

$A_1(s)$ to $A_N(s)$ and $B_1(s)$ to $B_N(s)$ in equation (28) are arbitrary functions of integration. λ_i , $\chi_i = (\chi_{1i}, \chi_{2i}, \dots, \chi_{Ni})^T$, respectively, stand for the i th eigenvalue and eigenvector of the characteristic matrix $\mathbf{M} = \bar{\kappa}^{-1}(s\bar{\mathbf{S}} + \mathbf{\Gamma})$ of the system of ODEs in equation (27), and $\mathbf{\Psi} = (\psi_1, \dots, \psi_N)^T = -s(s\bar{\mathbf{S}} + \mathbf{\Gamma})^{-1} \cdot \bar{\mathbf{a}}$ is its particular solution vector.

The condition of symmetry at the center axis of the problem geometry requires that $B_i(s) = 0$. The solution can be followed with the condition of drained boundary at $x = 1$ as follows:

$$p_i(1, t) = 0 \quad (29)$$

Substitution in equation (28) gives the following:

$$\begin{pmatrix} A_1(s) \cosh(x\sqrt{\lambda_1}) \\ \vdots \\ A_N(s) \cosh(x\sqrt{\lambda_N}) \end{pmatrix} = - \begin{pmatrix} \chi_{11} & \chi_{12} & \cdots & \chi_{1N} \\ \chi_{21} & \chi_{22} & \cdots & \chi_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \chi_{N1} & \chi_{N1} & \cdots & \chi_{NN} \end{pmatrix}^{-1} \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_N \end{pmatrix} \tilde{f} \quad (30)$$

or

$$A_i(s) = -\tilde{f} \frac{\text{Det}[\mathbf{X}_{\psi,i}]}{\text{Det}[\mathbf{X}] \times \cosh(x\sqrt{\lambda_i})} \quad (31)$$

Det in equation (31) denotes the determinant of a matrix. $\mathbf{X}_{\psi,i}$ is defined by replacing the i th column of the eigenvectors matrix $\mathbf{X} = [\chi_1^T, \chi_2^T, \dots, \chi_N^T]$ with Ψ . Substituting $A_i(s)$ back into equation (28) yields

$$\tilde{p}_i(r) = \tilde{f} \left[\psi_i - \frac{1}{\text{Det}[\mathbf{X}]} \sum_{j=1}^N \frac{\chi_{ij} \text{Det}[\mathbf{X}_{\psi,j}] \cosh(x\sqrt{\lambda_j})}{\cosh(\sqrt{\lambda_j})} \right] \quad (32)$$

The constitutive relations of a multiporoelastic material are rewritten below from equation (10)

$$\frac{\sigma_{ij}}{2G} = \varepsilon_{ij} + \left(\frac{\bar{v}}{1-2\bar{v}} \varepsilon - \gamma \sum_{i=1}^N \bar{\alpha}_i p_i \right) \delta_{ij} \quad (33)$$

where Lamé constant λ is replaced by its equivalent expression in terms of shear modulus G and Poisson's ratio ν and $\gamma = (2G \sum_{i=1}^N \bar{S}_i)^{-1}$. Equations (18) and (32) can be introduced into equation (33) to give expressions for stress tensor as follows:

$$\frac{\tilde{\sigma}_{ij}}{2G} = \varepsilon_{ij} - \left[\frac{\gamma(1-2\bar{v})}{(1-\bar{v})} \sum_{i=1}^N \bar{\alpha}_i \tilde{p}_i - \tilde{f} \left(\frac{\bar{v}}{1-2\bar{v}} \right) \right] \delta_{ij} \quad (34)$$

Thus far, Mandel's problem is reduced to finding the arbitrary function $f(t)$. The condition of normal stress at confined surface provides us with the required information for this purpose. It is easy to solve the stress equilibrium from equation (14) as follows:

$$\sigma_{xx} = 0 \quad (35)$$

which by substitution from equations (10) and (18) gives the following:

$$\varepsilon_{xx} = \frac{\sum_{i=1}^N \bar{\alpha}_i \tilde{p}_i}{\sum_{i=1}^N \bar{S}_i} - \tilde{f} \left(\frac{\bar{v}}{1-2\bar{v}} \right) \quad (36)$$

The plane strain state of the problem determines $\varepsilon_{zz} = \varepsilon - \varepsilon_{xx}$ as follows:

$$\tilde{\varepsilon}_{zz} = \tilde{f} \left(\frac{1-\bar{v}}{1-2\bar{v}} \right) \quad (37)$$

The vertical component of stress is consequently found from equations (32), (34), and (37) as

$$\tilde{\sigma}_{zz}(r) = \tilde{f} \left\{ \frac{1}{\text{Det}[\mathbf{X}]} \left[\sum_{i=1}^N \beta_i \text{Det}[\mathbf{X}_{\psi,i}] \frac{\cosh(r\sqrt{\lambda_i})}{\cosh(\sqrt{\lambda_i})} \right] + \psi \right\} \quad (38)$$

where

$$\beta_i = \frac{\sum_{j=1}^N \chi_{ji} \bar{\alpha}_j}{\sum_{j=1}^N \bar{S}_j} \quad (39)$$

$$\psi = 1 + 2a_1 - \frac{\sum_{j=1}^N \bar{\alpha}_j \psi_j}{\sum_{j=1}^N \bar{S}_j} \quad (40)$$

The boundary condition at $z=b$ reads

$$\int_0^1 \tilde{\sigma}_{zz} dx = -\frac{P_c}{s} \quad (41)$$

Table 1. The Intrinsic Properties of a GOM Shale

Porosity Level	Bulk Modulus	Poisson's Ratio	Void Ratio	Porosity	Permeability
Primary (Matrix)	$K_1 = 1.1$ GPa	$\nu_1 = 0.22$	$\nu_1 = 97\%$	$\phi_1 = 14\%$	$k_1 = 5$ nd
Secondary (Microfracture)	$K_2 = 55$ MPa	$\nu_2 = 0.22$	$\nu_2 = 2\%$	$\phi_2 = 95\%$	$k_2 = 5$ μ d
Tertiary (Macrofracture)	$K_3 = 22.5$ MPa	$\nu_3 = 0.22$	$\nu_3 = 1\%$	$\phi_3 = 95\%$	$k_3 = 5$ md

Substitution from equation (38) determines $\tilde{f}$ as follows:

$$\tilde{f} = - \frac{P_c \text{Det}[\mathbf{X}]}{s \left[\psi \text{Det}[\mathbf{X}] + \sum_{i=1}^N \frac{\beta_i \text{Det}[\mathbf{X}_{\psi,i}] \tanh(\sqrt{\lambda_i})}{\sqrt{\lambda_i}} \right]} \quad (42)$$

The pore pressure and vertical stress and strain would correspondingly be found as

$$\frac{\tilde{p}_i(r)}{P_c} = \frac{-\text{Det}[\mathbf{X}] \psi_i + \sum_{j=1}^N \chi_{ij} \text{Det}[\mathbf{X}_{\psi,j}] \frac{\cosh(r\sqrt{\lambda_j})}{\cosh(\sqrt{\lambda_j})}}{\gamma s \left[\psi \text{Det}[\mathbf{X}] + \sum_{i=1}^N \frac{\beta_i \text{Det}[\mathbf{X}_{\psi,i}] \tanh(\sqrt{\lambda_i})}{\sqrt{\lambda_i}} \right]} \quad (43)$$

$$\frac{\tilde{\sigma}(r)}{P_c} = - \frac{\text{Det}[\mathbf{X}] \psi + \sum_{i=1}^N \beta_i \text{Det}[\mathbf{X}_{\psi,i}] \frac{\cosh(r\sqrt{\lambda_i})}{\cosh(\sqrt{\lambda_i})}}{s \left[\text{Det}[\mathbf{X}] \psi + \sum_{i=1}^N \frac{\beta_i \text{Det}[\mathbf{X}_{\psi,i}] \tanh(\sqrt{\lambda_i})}{\sqrt{\lambda_i}} \right]} \quad (44)$$

$$\frac{2G\tilde{u}_z(b)}{P_c b} = - \frac{\text{Det}[\mathbf{X}]}{s \left[\psi \text{Det}[\mathbf{X}] + \sum_{i=1}^N \frac{\beta_i \text{Det}[\mathbf{X}_{\psi,i}] \tanh(\sqrt{\lambda_i})}{\sqrt{\lambda_i}} \right]} \quad (45)$$

Once the solution is found in transformed space, the inverse Laplace transformation can be taken to find expressions of the result functions in time domain. Because of the complex form of solution expressions, the numerical algorithm due to Stehfest [Stehfest, 1970; Cheng et al., 1994] has been employed for this purpose.

4. Numerical Example Solution

The obtained solution is used to simulate the response of triple-porosity shale comprising matrix pores and two fracture networks: the natural microfractures and macrofractures which may have been induced by tectonic activities or hydraulic fracturing. The set of physical properties from Gulf of Mexico (GOM) shale are adopted from Nguyen and Abousleiman [2010] with some modifications and extended to account for the tertiary porous network, which in this example represents shale's natural microfractures. These data are presented in Table 1. Also, the interporosity coefficients are given in Table 2. In many cases the secondary porosity plays the role of a mediator [Dehghanpour and Shirdel, 2011], that is, a big portion of fluid exchange between the matrix and larger fractures can take place only through microfractures. In order to simulate such condition, the exchange coefficient $\Gamma_{12} = \Gamma_{21} = 0$, while $\Gamma_{13}, \Gamma_{23} \neq 0$.

The grain's bulk modulus $K_s = 27$ GPa. The fluid's Bulk modulus and viscosity are $K_f = 1744$ GPa and $\mu = 1$ cp, respectively. The specimen's dimension in the direction of fluid flow $a = 3$ cm is taken in accord with laboratory condition of related test setup.

Should the void ratio of either higher-permeability porosity network be allowed to vanish, the multiple-porosity/multiple-permeability solution renders the corresponding lower porosity level responses of the

Table 2. The Interporosity Exchange Coefficients (MPa s^{-1})

	Matrix	Macrofracture	Microfracture
Matrix	—	0	1.67×10^{-16}
Macrofracture	0	—	1.67×10^{-18}
Microfracture	1.67×10^{-16}	1.67×10^{-18}	—

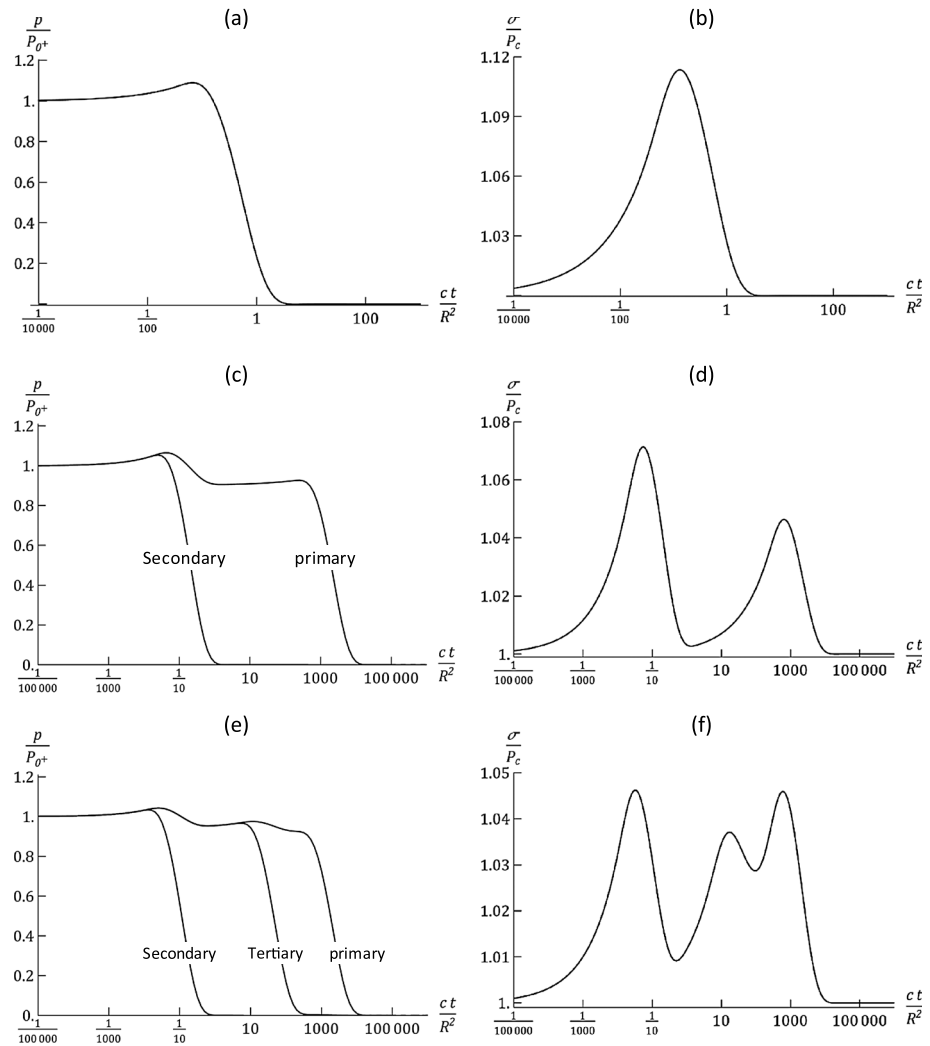


Figure 5. Pore pressure p and stress σ distributions obtained for solutions to Mandel's problem with (a, b) single-, (c, d) double-, and (e, f) triple-porosity networks.

material. For example, letting $v_2 = v_3 = 0$ in Table 1 should recover the single-porosity solution related to the shale matrix only. This result can be obtained by applying the following replacements and substitutions in equations (43)–(45)

$$N \rightarrow 1; \quad \mathbf{S} = S; \quad \mathbf{\Gamma} = 0; \quad \kappa = \kappa; \quad \mathbf{X} \rightarrow 1; \quad \lambda \rightarrow s; \quad \Psi \rightarrow -\alpha$$

The results are shown in Figures 5a and 5b, where the pore pressure p as well as confining stress σ at the center point or axis of all geometries are plotted versus time. The solution exhibits a single time scale $t_0 = a^2/c$, where $c = k/(\mu S)$ with the Mandel-Cryer effect captured as the pressure and stress override above P_0^+ and P_c at early times.

The initial pore pressure can be obtained from the initial value theorem of Laplace transform as follows:

$$p_0^+ = \frac{B(1 + v_u)}{3} \quad (46)$$

where v_u is the undrained Poisson's ratio. The confining stress σ_i in each dimension would cause an initial pressure buildup of $p_0^+ i = (B/3)\epsilon_i$. We notice the presumed condition of plane strain in rectangular Mandel's problem, which implies that $\sigma_{yy} = -v_u \sigma_{xx}$. Therefore, the initial pressure rise would be found as the expression shown in equation (46).

A dual-porosity model of the shale sample may be recovered by letting $v_3 = 3\%$, while still assuming that $v_2 = 0$. Correspondingly, the following replacements and substitutions should be made in equations (43)–(45)

$$N \rightarrow 2; \quad \mathbf{S} \rightarrow \begin{pmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{pmatrix}; \quad \mathbf{\Gamma} \rightarrow \mathbf{\Gamma}_{12} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}; \quad \mathbf{\kappa} \rightarrow \begin{pmatrix} \kappa_1 & 0 \\ 0 & \kappa_2 \end{pmatrix}; \quad \mathbf{\Psi} \rightarrow - \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}.$$

Results are shown in Figures 5c and 5d, where the solution exhibits a dual-time scale behavior. However, unlike the single-porosity case, these time scales are attributed to the eigenvalues of the characteristic equation of solution partial differential equations and are themselves time-dependent variables. Also, we notice that the portion of rock's response corresponding to higher-permeability network (secondary porosity or fractures) would, in general, affect and interact with the one related to lower permeability (primary porosity or matrix). This occurrence is due to the coupled deformation processes, as well as the interporosity fluid exchange effects. For this reason, the curves corresponding to pore pressure in shale's matrix exhibit a dual Mandel-Cryer effect. Also, the initial pressure both in fracture p_{20^+} and matrix p_{10^+} would be less than the initial pressure rise of the single-porosity solution due to larger compressibility of the dual-porosity mixture compared to the single-porosity matrix.

We may further relax the condition of dual-porosity flow by allowing $v_3 = 1\%$ and $v_2 = 2\%$, as shown in Table 1. In such a case, a triple-porosity/triple-permeability solution for response of the rock is attained. This result can be obtained by applying the following replacements and substitutions in equations (43)–(45)

$$N \rightarrow 3; \quad \mathbf{S} \rightarrow \begin{pmatrix} S_{11} & S_{12} & S_{13} \\ S_{12} & S_{22} & S_{23} \\ S_{13} & S_{23} & S_{33} \end{pmatrix}; \quad \mathbf{\Gamma} \rightarrow \begin{pmatrix} \Gamma_{12} + \Gamma_{13} & -\Gamma_{12} & -\Gamma_{13} \\ -\Gamma_{12} & \Gamma_{12} + \Gamma_{23} & -\Gamma_{23} \\ -\Gamma_{13} & -\Gamma_{23} & \Gamma_{13} + \Gamma_{23} \end{pmatrix};$$

$$\mathbf{\kappa} \rightarrow \begin{pmatrix} \kappa_1 & 0 & 0 \\ 0 & \kappa_2 & 0 \\ 0 & 0 & \kappa_3 \end{pmatrix}; \quad \mathbf{\Psi} \rightarrow - \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}.$$

Results are shown in Figures 5e and 5f, where the solution exhibits a triple-time scale and Mandel-Cryer effect.

5. Conclusions

This study offered an extension to the existing theories of single- and dual-porosity, fluid-saturated, and elastic media to account for higher levels of pore structure complexity, namely, the multiple-porosity and multiple-permeability poroelasticity. Illustrative and elaborate presentations of the conceptual models provided in this paper attempted to explore the fundamental concepts underlying a multiple-poroelastic response to external tractions and further to develop their generalized constitutive relations. In particular, the Appendix A presents a closure solution to the problem of material property coefficients of a multiple-porosity mixture, whose constituents follow Biot's formulations of stress-strain relation.

A complete analytical solution to Mandel's problem of multiple poroelasticity is developed. Results indicate that the problem's time scales, and consequently, the associated coupled phenomenon, i.e., Mandel-Cryer effect, would proliferate when the number of distinct porosity networks in the analytical model is increased. Neglecting or oversimplifying the effects of distinguishable pore structures within the specimen's matrix might produce errors which often tend to oversimplify the trends of variation of either flow or deformation aspects of its intricate response to application of stress or pore pressure disturbances.

Appendix A: Mechanical Property Coefficients of a Multiple-Porosity Mixture

Consistent and complete mechanical characterization of multiple-porosity mixtures is still an open problem in poromechanics society [Berryman, 2002]. This problem is illustrated in Figure 6, where the coefficient

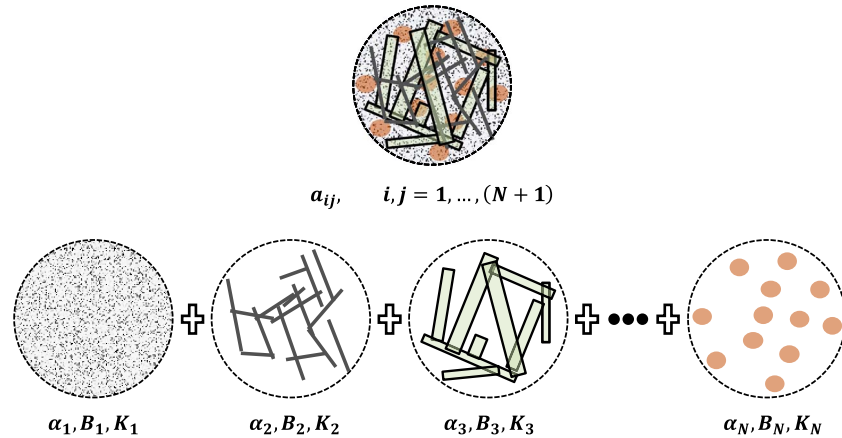


Figure 6. N poroelastic constituents characterized by α_i, B_i, K_i comprising an N porosity poroelastic mixture with mechanical properties a_{ij} .

parameters a_{ij} of an N porosity mixture, as shown in equation (7), are to be determined from the intrinsic properties of its individual porous components.

This objective inherently incorporates the assumption that the individual components are *Gassmann materials*, that is to say, the response of each poroelastic component to deformation follows equation (1) as rewritten below

$$\begin{pmatrix} \varepsilon_i \\ \zeta_i/v_i \end{pmatrix} = \frac{1}{K} \begin{pmatrix} 1 & \alpha_i \\ \alpha_i & \alpha_i/B_i \end{pmatrix} \begin{pmatrix} \sigma_i \\ p_i \end{pmatrix}, \quad i = 1, \dots, N \quad (\text{A1})$$

Addition of the volume fraction v_i in equation (A1) accounts for definition of ζ_i in terms of the unit volume of the i th porosity network.

A1. Dual-Porosity Poroelastic Mixture

Berryman [2002] extended the theory of single-porosity poromechanics to the case of dual-porosity mixtures by suggesting the constitutive relation displayed in equation (4) rewritten below

$$\begin{pmatrix} \varepsilon \\ \zeta_1 \\ \zeta_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} \sigma \\ p_1 \\ p_2 \end{pmatrix} \quad (\text{A2})$$

The study suggested that parameters a_{ij} can be found by running a thought experiment in which the mixture undergoes a process of *self-similar* expansion. That is to say, a state of confining stress σ , together with two pore pressures p_1, p_2 , which retain equal and uniform volumetric strain in the individual porous components and hence in the overall mixture, i.e., $\varepsilon = \varepsilon_1 = \varepsilon_2$. Through a set of relations obtained by introducing equations (A1) and (A2) into this condition, the expressions for a_{ij} were recovered in terms of α_i, B_i, K_i .

Here we follow Berryman's method for dual-porosity materials by using general manifestations of σ, p_1, p_2 . Figure 6 shows the schematic of a dual-porosity matrix which undergoes an external confining pressure σ and the two pore pressures p_1 and p_2 , as well as the combinations of opening or closing drainage vents w_1 and w_2 .

A selection of the dashed line or solid line at the interface of the two components, respectively, implies a pervious or impervious boundary. The classical undrained compression test, for instance, can be obtained by setting $p_1 = p_2 = p$ and $\sigma \neq 0$ when both vents w_1, w_2 are closed and the boundaries of the constituents are allowed to be pervious.

It should be noted that, however, not necessarily all experiments conceivable through the setup shown in Figure 7 would generate linearly independent equations. Here we consider the most general case in which

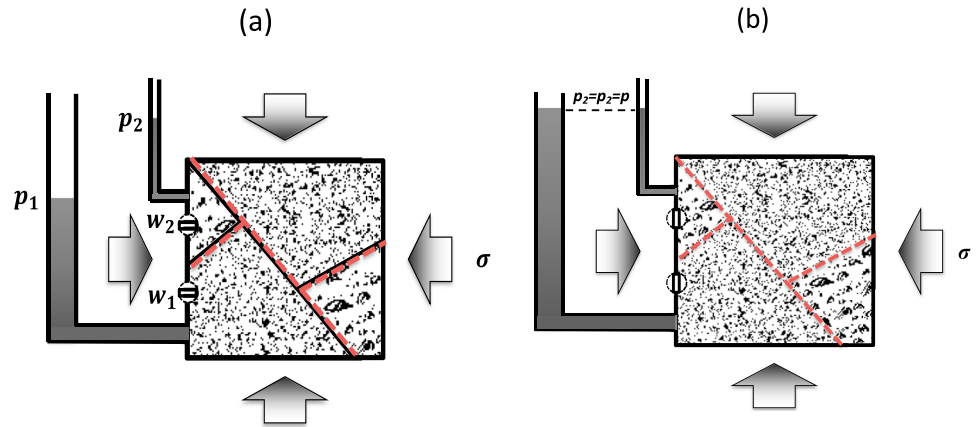


Figure 7. (a) Schematics of the dual-porosity model at microscopic level of investigation, with illustrations of the two imaginary vents w_1 and w_2 , along with the pervious (dashed) or impervious (solid) boundaries between the porous constituents. (b) Schematics of the undrained confining test demonstrated by closed vents and pervious interporosity boundary.

the controlling variables σ, p_1, p_2 can be adjusted independently. In order to maintain the distinction between the two pore pressures, a condition of impervious interporosity boundary should be retained, while both vents w_1, w_2 are kept closed. Under such conditions, equation (A1) provides the following set of relations for the two porous constituents

$$\epsilon_i = -\frac{1}{K_i} \sigma_i - \frac{\alpha_i}{K_i} p_i, \quad i = 1, 2 \quad (\text{A3})$$

$$\frac{\zeta_i}{v_i} = -\frac{\alpha_i}{K_i} \sigma_i + \frac{\alpha_i}{K_i B_i} p_i, \quad i = 1, 2 \quad (\text{A4})$$

Likewise, equation (A2) results in the following relations for the overall mixture

$$\epsilon = a_{11} \cdot \sigma - a_{12} p_1 - a_{13} p_2 \quad (\text{A5})$$

$$\zeta_1 = a_{12} \cdot \sigma + a_{22} p_1 + a_{23} p_2 \quad (\text{A6})$$

$$\zeta_2 = a_{13} \cdot \sigma + a_{23} p_1 + a_{33} p_2 \quad (\text{A7})$$

We also notice the conditions of compatibility in volumetric strain of mixture's solid skeleton and the fluid content increment

$$\epsilon = v_1 \epsilon_1 + v_2 \epsilon_2 \quad (\text{A8})$$

$$\zeta = \zeta_1 + \zeta_2 \quad (\text{A9})$$

Lastly, we consider the condition of microscopic equilibrium of a material element encompassing the constituents' interface. In order for this element to be statically in equilibrium, we must have $\sigma_1 = \sigma_2 = \sigma$. This would be a valid solution of the equilibrium equations and boundary conditions.

After some rearrangement of the terms and expressions, equations (A3)–(A9) are simplified to

$$\left(1 + \frac{a_{12} K_1}{v_1 \alpha_1}\right) \sigma - \left(\frac{1}{B_1} - \frac{a_{22} K_1}{v_1 \alpha_1}\right) p_1 + \frac{a_{23} K_1}{v_1 \alpha_1} p_2 = 0 \quad (\text{A10})$$

$$\left(1 + \frac{a_{13} K_2}{v_2 \alpha_2}\right) \sigma + \frac{a_{23} K_2}{v_2 \alpha_2} p_1 - \left(\frac{1}{B_2} - \frac{a_{33} K_2}{v_2 \alpha_2}\right) p_2 = 0 \quad (\text{A11})$$

$$\left(a_{11} + \frac{a_{12}}{\alpha_1} + \frac{a_{13}}{\alpha_2}\right) \sigma + \left[a_{12} + \frac{a_{22}}{\alpha_1} + \frac{v_1(1 - B_1 \alpha_1)}{B_1 K_1}\right] p_1 - \left[a_{13} + \frac{a_{33}}{\alpha_2} + \frac{v_2(1 - B_2 \alpha_2)}{B_2 K_2}\right] p_2 = 0 \quad (\text{A12})$$

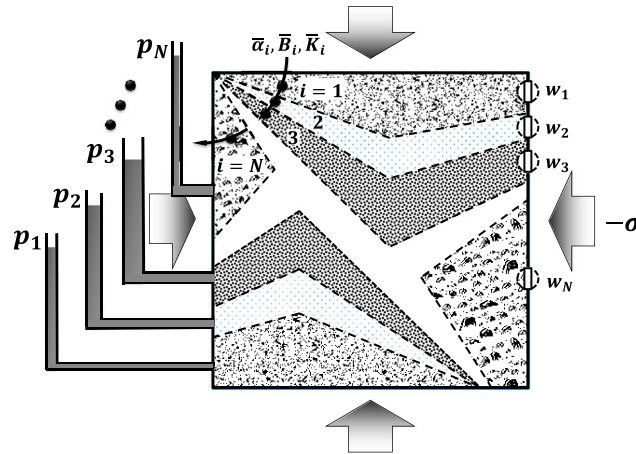


Figure 8. Conceptual representation of the multiple-porosity model at the microscopic level of investigation, with illustrations of N imaginary vents $w_1, \dots, w_N$.

Since σ, p_1, p_2 are selected arbitrarily, their coefficients in equations (A10, A11, A12) should vanish. Therefore, the material properties a_{ij} are found as follows:

$$a_{11} = \frac{1}{K} = \frac{v_1}{K_1} + \frac{v_2}{K_2} \quad (\text{A13a})$$

$$a_{12} = -\frac{v_1 \alpha_1}{K_1} \quad (\text{A13b})$$

$$a_{12} = -\frac{v_2 \alpha_2}{K_2} \quad (\text{A13c})$$

$$a_{22} = \frac{v_1 \alpha_1}{B_1 K_1} \quad (\text{A13d})$$

$$a_{33} = \frac{v_2 \alpha_2}{B_2 K_2} \quad (\text{A13e})$$

$$a_{23} = 0 \quad (\text{A13f})$$

The set of equations (A13) completes the poroelastic characterization of a dual-porosity mixture. These results are identical with derivations of *Berryman* [2002] and *Berryman and Pride* [2002] if the effective bulk modulus of the mixture in their derivations is taken as the harmonic average of the constituents' compressibilities.

A2. Multiple-Porosity Poroelastic Mixture

The procedure for finding poroelastic coefficients of a multiple-porosity material follows and extends the formulations of previous section. Similar to illustrations of Figure 7a, Figure 8 presents the setup for a group of thought experiments on the multiple-porosity model at microscopic level, where N imaginary vents $w_1, \dots, w_N$ are allowing for drained or undrained condition of confinement for individual porosity networks.

Again, we consider the most general case in which the controlling parameters $\sigma, p_1, \dots, p_2$ can be adjusted independently. The constitutive relations for individual components of the composite for such experiment can be written as

$$\epsilon_i = -\frac{1}{K_i} \sigma_i - \frac{\alpha_i}{K_i} p_i, \quad i = 1, \dots, n \quad (\text{A14})$$

$$\frac{\zeta_i}{v_i} = -\frac{\alpha_i}{K_i} \sigma_i + \frac{\alpha_i}{K_i B_i} p_i, \quad i = 1, \dots, n \quad (\text{A15})$$

Similarly, the following equations can be written for the bulk material

$$\epsilon = a_{11}\sigma - \sum_{k=2}^n a_{1k} p_k \quad (\text{A16})$$

$$\zeta_{(i-1)} = a_{i1}\sigma - \sum_{k=2}^n a_{ik} p_{(i-1)}, \quad i = 2, \dots, n+1 \quad (\text{A17})$$

The geometrical compatibility can be stated as follows:

$$\sum_{k=1}^n v_k \epsilon_k = \epsilon \quad (\text{A18})$$

Substitution of equations (A14) and (A18) in equation (A16) yields

$$\left(a_{11} - \sum_{k=1}^n \frac{v_k}{K_k} \right) \sigma + \sum_{i=2}^{n+1} \left(a_{1i} + \frac{v_i \alpha_i}{K_i} \right) p_{(i-1)} \quad (\text{A19})$$

The condition of the static equilibrium of components $\sigma_i = \sigma$ is used here as before. Since σ, p_i are selected arbitrarily, their coefficients in equation (A19) should vanish. That would determine the coefficients a_{1i} as follows:

$$a_{11} = \sum_{k=1}^n \frac{v_k}{K_k} \quad (\text{A20})$$

$$a_{1i} = -\frac{v_i \alpha_i}{K_i}, \quad i = 1, \dots, n+1 \quad (\text{A21})$$

We may proceed with equation (A17), which after some arrangement of the expressions returns

$$\left(a_{1i} + \frac{v_i \alpha_i}{K_i} \right) \sigma + \left(a_{ii} - \frac{v_i \alpha_i}{B_i K_i} \right) p_i + \sum_{k=2, k \neq i}^{n, k \neq i} a_{ik} p_i = 0 \quad (\text{A22})$$

Again, since σ, p_i are selected arbitrarily, their coefficients in equation (A22) should vanish. The first term reproduces equation (A21), while the second and third terms yield

$$a_{ij} = 0, \quad (i > 2, i \neq j) \quad (\text{A23})$$

$$a_{ii} = \frac{v_i \alpha_i}{B_i K_i}, \quad i = 2, \dots, n+1 \quad (\text{A24})$$

Equations (A20), (A21), (A23), and (A24) complete the information required to determine the mechanical properties coefficients for a multiple-porosity homogenous poroelastic material. These equations can be represented in matrix form as follows:

$$\begin{pmatrix} \epsilon \\ \zeta_1 \\ \vdots \\ \zeta_N \end{pmatrix} = \begin{pmatrix} \sum_{k=1}^n \frac{v_k}{K_k} & -\frac{v_1 \alpha_1}{K_1} & \dots & -\frac{v_N \alpha_N}{K_N} \\ -\frac{v_1 \alpha_1}{K_1} & \frac{v_1 \alpha_1}{B_1 K_1} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{v_N \alpha_N}{K_N} & 0 & \dots & \frac{v_N \alpha_N}{B_N K_N} \end{pmatrix} \begin{pmatrix} \sigma \\ p_1 \\ \vdots \\ p_N \end{pmatrix} \quad (\text{A25})$$

The elements of the coefficient matrix in equation (A25) may be interpreted through correspondence with single-porosity poroelastic properties as follows: a_{11} is the mixture's effective compressibility, which is derived as the weighted average of the components' compressibilities. This result is consistent with conventional averaging methods for the compressibility of mixtures. The reminder elements in first row of coefficient matrix represent the generalized Biot-Willis coefficients. It should be noted that none of these elements represent the effective Biot-Willis coefficient of the whole system. In order to find such property,

we examine the undrained thought experiment, as shown in Figure 7b. We can define the overall undrained bulk modulus of a multiporosity system as

$$\frac{1}{K_u} = \frac{\partial e}{\partial \sigma} \Big|_{\zeta=0} = \frac{\partial e}{\partial \sigma} + \frac{\partial e}{\partial p} \frac{\partial p}{\partial \sigma} \Big|_{\zeta=0} = a_{11} + \left(\sum_{j=2}^{n+1} a_{1j} \right) \bar{B} \quad (\text{A26})$$

where $\bar{B}$ is the overall Skempton's pore pressure coefficient. Equation (A26) can be rearranged to give the overall Biot-Willis coefficient as

$$\bar{\alpha} = \left(\frac{1 - K/K_u}{B} \right) = - \sum_{j=2}^{n+1} a_{1j} \quad (\text{A27})$$

We notice that all other off-diagonal elements are obtained to be zero. Derivations of *Berryman and Pride* [2002] and *Berryman* [2002] do not identically coincide this result. However, discussions within same publications confirm that although nonvanishing, their magnitude would be practically insignificant. For further reading on this issue, see variations of a_{23} in Figure 3 and discussions on p. 845 of *Berryman* [2002], as well as discussions on p. 10 and variations of a_{23} in Figure 4 of *Berryman and Pride* [2002]. In retrospect, results of current analysis suggest that these terms would vanish identically in order to maintain the microscopic static equilibrium of the mixture's constituents and at the same time satisfy the corresponding constitutive relations.

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